



Module 01: PRAHARI

Lectures Stakeholder briefing · Amit Dua, Arnav, Aria · September 2026

1.1 Who this document is for

This is the teaching note for PRAHARI. Readers are Home Department evaluators, departmental operators, and the student team. The voice is a lecturer explaining a machine to a bright ten-year-old, then naming the real parts so an adult can inspect them.

Every number is one of three labels. MEASURED is a fact from this laptop or the Sentinel sandbox. DESIGN TARGET is statewide arithmetic we have not run at 80,000 cameras. CONJECTURED is unused in this note.

1.2 The problem, in one picture

Imagine 26 schools in one city. Each school has cameras. Each school bought a different cupboard to store the tapes. Some cupboards keep tapes for 7 days. Some keep them for 15. The head office has a notebook of stolen bicycles and missing children. Nobody in the camera room opens that notebook when a bicycle rolls past.

That is Gujarat's CCTV estate. Twenty-six departments. Analog and IP cameras. Sites from border districts to Valsad, Dahod, Somnath, Jamnagar, and Dwarka. Watchlists already exist: VAHAN, SARTHI, eGujCop, AFIS, NAFIS. The cameras do not match against those lists. There is no statewide census of which public-domain cameras exist, whether they are alive, or where coverage stops.

The official page is <https://sentinel.gujarat.gov.in/problems>. Phase 1 closes 07 September 2026 at 12:00 IST. PRAHARI is Category 1 (student). The team is Amit Dua, Arnav, and Aria.

Key idea

The State does not need a new cupboard for every tape. It needs one notebook that watches the cameras it already paid for.

The official core goal is a secure, scalable, interoperable, technically feasible, and cost-effective approach that uses existing infrastructure to the maximum practical extent. A student PoC cannot replace 26 VMS contracts. A central recorder that ingests 80,000 full-HD streams (DESIGN TARGET) for 15 days needs statewide dark fibre, petabytes of hot storage, and a GPU farm this team does not have. PRAHARI therefore adds an intelligence plane on top of the systems that already record video.

1.3 The four official models, and the choice

The official page lists four reference models and permits a hybrid. Think of them as four ways to organise a school.

Definition 1.3.1 (Model 1: registry and GIS). A roll call. Who owns the camera, where it sits, whether it is alive, how many days it keeps tape. No live picture is required.

Definition 1.3.2 (Model 2: unified viewing). One window that can open many classrooms. The school keeps its own cupboard. The window speaks RTSP, ONVIF, or a vendor API. ANPR may run on a sampled frame without storing every frame.

Definition 1.3.3 (Model 3: VMS federation). A post office between cupboards. Each vendor gets an adapter. Events travel on a bus. The original cupboard stays.

Definition 1.3.4 (Model 4: central VMS). One giant cupboard for the whole city. Ingest, store, playback, full AI, hot/warm/cold tape. This is a rip-and-replace.

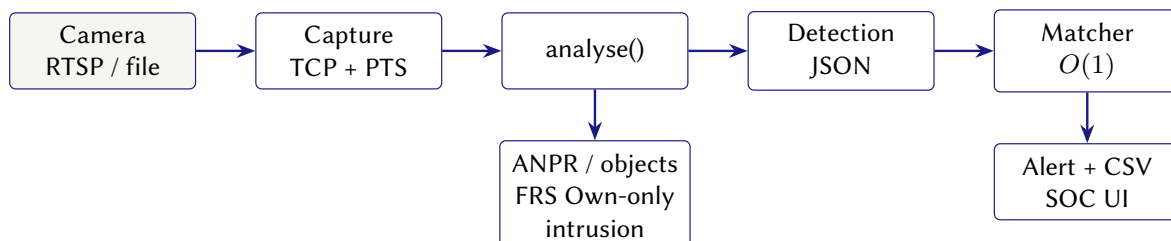
Options we had		
Option	We take it?	Why
Model 1 only	No	A roll call does not catch a stolen car.
Model 2 only	No	Viewing without a watchlist fails the test case.
Model 3 full SDK zoo	No	Seven days cannot wrap every vendor SDK.
Model 4 statewide	No	We cannot host 80,000 recordings (DESIGN TARGET).
Hybrid 1+2+thin 3	Yes	Census now, viewing now, events now, recording later.

Definition 1.3.5 (PRAHARI hybrid). Model 1 supplies camera metadata, GIS, health, and gap reports. Model 2 opens reachable RTSP, HLS, WHEP, or ONVIF streams and runs analytics on sampled frames. Thin Model 3 is a detection-event JSON bus, an in-memory watchlist matcher, and GET /api/track/{plate}. Model 4 central recording is Phase-2 for selected cameras only.

A new vendor is one adapter. The detection JSON does not change if the bus later becomes Kafka. Kafka, Kubernetes, and Ceph are named as later swap points. They are not in this PoC.

1.4 How a picture becomes a red card

A ten-year-old can follow this path. An adult can grep it.



1. A camera produces a live stream, or a file clip for the own-feed demo.

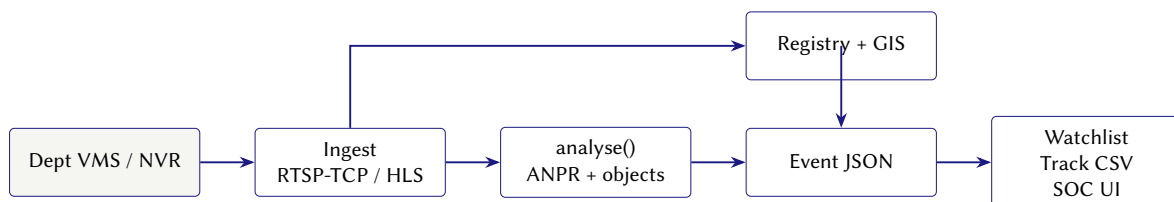


2. Capture opens the stream over RTSP-TCP (or HLS, or a file). Time comes from PTS, not the wall clock.
3. analyse() looks at about one frame per second.
4. Each hit becomes a detection JSON row in SQLite.
5. The matcher checks a notebook (the representative watchlist) in $O(1)$.
6. A stolen plate raises a CRITICAL alert. The SOC tab turns red. A CSV records timestamps.

The stolen-car story

The notebook has GJ01AB1234 marked STOLEN (WL-001). Six seed cameras already saw that plate from Valsad to Gandhinagar. On 03 September 2026 an operator confirm on Paldi Circle (cam04) appended a seventh live point. The six seed ids stayed. The CSV is the evaluator artefact.

1.5 Architecture of the PoC



The PoC runs FastAPI and SQLite on one laptop. The UI is Leaflet plus vanilla JS on port 8080. Capture uses OpenCV with FFmpeg.

PoC versus later swap

Piece	PoC	Later swap
API	FastAPI + Uvicorn	same HTTP contract
UI	Leaflet + vanilla JS	React if a wall needs it
Store	SQLite	PostgreSQL + PostGIS
Capture	OpenCV + FFmpeg, RTSP TCP	DeepStream on regional GPUs
ANPR	morphology + Tesseract	YOLO plate detector + PaddleOCR
Objects	CPU blob / OpenCV DNN	YOLO11 at 1 fps on regional GPU
FRS	enrolled gallery, Own cameras	dedicated cameras, human confirm
Alerts	in-process WebSocket	Redis / Kafka
Auth	HTTP Basic + signed cookie	department SSO

1.6 Registry and GIS (Model 1)

A roll call needs a row per camera: id, department, ownership, consent, lat, lon, protocol, retention, health. Three doors put a row in: a CSV, a form, a REST call. A fourth door pulls the Sentinel catalogue.

Health is a colour on the Gujarat map. Three failed probes turn a pin red. Dahod GSRTC is seeded offline on purpose so a gap report has something true to say. Food and Civil Supplies cameras with 7-day retention appear in the short-retention list.



Private-permitted mall cameras require `consent=true`. A Home department viewer never sees that mall. That is the law of the notebook, not a CSS trick.

The live Sentinel catalogue has no coordinates. Those rows therefore do not sit on the map (the map already skips `lat = 0, lon = 0`). Operators open `cam04` from the Cameras table. The UI states that fact in words.

1.7 Live capture and the integrator laws

The sandbox publishes each camera as a live RTP/RTSP stream. One second of video takes one second to arrive. There is no rewind and no file download.

On 03 September 2026 the live portal is `https://cctv.corp8.cloud/`. Catalogue `GET /cameras.json`, after password login, returns a JSON array of `{id, name}`. Thirty cameras were listed. `/api/ingest` returns HTTP 404 on this host.

HLS playlists live at `https://cctv.corp8.cloud/<id>/index.m3u8`. The origin wants the access cookie and a browser User-Agent. RTSP is not on the TLS hostname. The live resource page publishes `rtsp://103.250.160.189:8554/stream/<id>`.

Warning

`/stream/<id>` answers range requests for a player. `curl` or `wget` of that path yields a partial file that looks complete. A team that ships that file as the government-feed demo has not consumed the grid. PRAHARI consumes live. It does not publish to the gateway.

Why TCP, not UDP? UDP dies across NAT and most campus firewalls. Partial UDP looks like a broken model. The capture client sets `OPENCV_FFMPEG_CAPTURE_OPTIONS=rtsp_transport;tcp` before it imports OpenCV. If port 8554 is blocked, the client falls back to HLS.

Why PTS, not the wall clock? After connect, the gateway may replay a group of pictures faster than real time. A tracker that stamps arrival time invents impossible speeds. Event time is `CAP_PROP_POS_MSEC`. Declared FPS is unused.

Why backoff 2 s to 30 s? Feeds restart. A tight reconnect loop is a denial of service against ourselves.

Why a scene cut? Each sandbox feed loops. At the loop the picture jumps. Object track ids and face tracks reset. A PTS rewind or a gap above 5 s is that jump.

Implementation

`tests/test_integrator_laws.py` greps the capture client for the TCP flag, forbids `CAP_PROP_FPS` in services, and forbids publish helpers. `test_scene_cut.py` checks rewind and gap, and checks that object track ids reset.

1.8 analyse(): one door for every engine

A child should not have four different “look at this picture” buttons that disagree. Adults should not either.

Definition 1.8.1 (`analyse(frame, camera, pts_ms)`). The only worker entry. It returns a list of detection dicts. It does not insert them. The ingest router writes crops, inserts rows, and calls the matcher.



Default engines are anpr, objects. Own cameras may also list faces. Government catalogue ids matching cam plus digits never receive the faces engine. Ownership Gov never receives it either. The log line is frs_refused.

Godown camera CAM-FCS-001 additionally runs intrusion: a person box that overlaps a stored ROI by at least 30 percent of the person area.

Plate fields stay frozen so old callers keep working. Additive fields are entity_type, entity_id, face_id, object_class, bbox, track_id, source.

1.9 ANPR: options and the choice

A number plate is a name tag glued to a car. We need the letters in a standard spelling so GJ 01 AB 1234 and GJ-01-AB-1234 are the same name.

Definition 1.9.1 (Indian plate normaliser). After stripping spaces and hyphens, the string must match `^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{3,4}$`. Anything else is dropped.

ANPR options		
Option	Taken?	Constraint
Train a detector from scratch	No	Seven days, no labelled Gujarat set.
Require a GPU YOLO now	No	The laptop has GPU count MEASURED 0.
Tesseract + morphology	Yes	CPU, swap behind recognize().
Operator confirm	Yes	A readable plate cannot fail a live demo.

`recognize(frame_bgr)` returns {plate, plate_raw, confidence, crop}. Callers never import Tesseract. If the Tesseract binary is missing, the function returns an empty plate. It does not crash. `ANPR_ENGINE=yolo` is import-guarded and unused unless weights exist.

Operator confirm writes confidence 1.0 and `source=operator_confirm`. A confirm row is never labelled `source=anpr`. That honesty is a test, not a slogan.

MEASURED 03–04 September 2026: Tesseract was not on PATH. Confirm on cam04 wrote GJ01AB1234. Eight own-feed stills at PTS 1 s produced no plate. The log says so.

1.10 Object detection: options and the choice

A person is a blob that looks like a person. A car is a blob that looks like a car. The official evaluation scores person and vehicle detection, not only plates.



Object-engine options		
Option	Taken?	Constraint
YOLOv8n as the only path	No	Extra dependency, GPU optional, tests would skip.
OpenCV DNN with vendored weights	Later	Weights are large. Tests must not need a download.
Skin-tone blob fallback	Yes for tests	A drawn rectangle is a person if it occupies $\geq 8\%$ of the frame.
Skip objects entirely	No	Eval area 05 names them.

Classes kept: person, car, motorcycle, bus, truck, bicycle. Others drop. Each box gets a per-camera track_id. A scene cut resets those ids. Object rows do not raise a watchlist card unless they are wrapped as intrusion.

GET /api/objects/report.csv is the evaluator artefact: timestamp, camera, class, confidence.

1.11 Lawful FRS: options and the choice

A face is a name tag a person did not pin on. Matching faces of strangers on a public street is a rights problem, not a prize feature.

FRS options		
Option	Taken?	Constraint
No FRS at all	No	HLD text and eval 05 name recognition.
Wild FRS on Paldi Circle	No	Unknown members of the public. Unlawful in this PoC.
Live AFIS / NAFIS pipe	No	We have no ministry API. Claiming one is a false claim.
Enrolled gallery, Own cameras	Yes	Consented adults or synthetic fixtures. Human confirm.

Warning

The faces engine is refused when ownership is not Own, and when camera_id matches cam plus digits. Tests post a face-like still to cam04 and assert zero person events. The government-feed demo never shows FRS.

The gallery stores crops under data/faces/{gallery_id}/. Match uses a colour histogram of a 64×64 crop (LBPH if OpenCV contrib is present). GET /api/faces/gallery returns ids and counts. It does not return embeddings. WL-004 is a person row. The matcher fires on face_id, not on an empty plate.

Synthetic fixtures are geometric ellipses with unique noise. Tests do not need a photograph of a real person. A human demo may add two consented adult photos. Never a minor. Never a crop from cctv.corp8.cloud.

1.12 Intrusion on a godown



Food and Civil Supplies cameras watch a warehouse, not a highway. A person standing in a drawn rectangle is an intrusion.

The ROI lives in `cameras.extra_json`. Overlap of at least 30 percent of the person box with the ROI raises `entity_type=intrusion`, category `INTRUSION`, priority `CRITICAL`. Two hits within 120 s on the same camera increment a counter. There is no fourth neural net. Intrusion wraps object boxes.

1.13 Watchlist, alerts, track

The notebook is a representative watchlist. It is not a live ministry pipe. Categories include `STOLEN`, `BLACKLIST`, `OBSERVE`, and a person row `MISSING_ASSOCIATE`.

The matcher keeps plates in a dictionary and gallery ids in another. Lookup is $O(1)$. Reload is on demand (and on boot). Duplicate suppression: the same entity on the same camera within 120 s increments counter. Priority `CRITICAL` is stolen, wanted, or intrusion. `HIGH` is blacklist or the person associate. `LOW` is log.

WL-001 cannot be deleted. That is the seeded evaluation plate.

Designated vehicle GJ01AB1234

GET `/api/track/GJ01AB1234` returns the six seed points in this order: CAM-VAL-001 Valsad 06:12, CAM-SUR-001 Surat 07:48, CAM-AHD-004 Narol-Naroda 09:05, CAM-AHD-001 SG Highway 09:22, CAM-GNR-003 Koba 09:51, CAM-GNR-001 Gandhinagar Sector 21 10:04. Live confirm on `cam04` appended. CSV columns include plate, camera, timestamp, location.

Cross-camera tracking in this PoC is plate sightings joined to camera GIS. It is not a Re-ID embedding across Paldi Circle. A teleport flag fires if implied speed exceeds 180 km/h. That flag is information, not a hard fail.

Alerts push on WebSocket `/ws/alerts`. The UI also polls every 5 s so a blocked socket does not hide a `CRITICAL` card. Person cards show a name or gallery id, not a blank plate.

1.14 Security and privacy

Roles: `superadmin`, `soc_operator` (judge), `dept_viewer`, `auditor`. A Home viewer sees Home cameras only. An auditor cannot POST. Stream tiles use a 60-second HMAC token. Responses to the browser omit `rtsp://`. The Sentinel cookie stays on the server.

Path jail keeps crops inside the media root. HLS origin is pinned to the configured Sentinel host. Passwords compare with `hmac.compare_digest`.

Audit rows cover onboard, watchlist edit, ack, report download, plate confirm, and face confirm. Stream URLs are not stored in audit detail.

1.15 Scale and cost

A ten-year-old version: do not ship every school video to one room. Ship a tiny still once a second, and only from cameras that matter. The wall shows 16 to 64 live tiles. The rest is metadata.



Numbers below mix MEASURED laptop facts with DESIGN TARGET statewide arithmetic.

Assume 45,000 public-domain cameras run analytics at 1 fps with an 80 KB JPEG crop (DESIGN TARGET). Naive centralisation is $45,000 \times 80 \text{ KB/s} \approx 3.6 \text{ GB/s}$. Five regional GPU sites (Ahmedabad, Surat, Rajkot, Vadodara, Bhuj) cut that to about 720 MB/s per site. Low-connectivity outposts send events only.

MEASURED on 04 September 2026 on this laptop: four open file captures succeed; the fifth is rejected. Mean JPEG from eight own-feed stills is 41 KB. `/api/health p99` is 9.4 ms over 50 calls. GPU count is 0. Statewide 80 KB and regional GPUs remain DESIGN TARGET.

Order-of-magnitude yearly cost of the intelligence plane, not of replacing 26 VMS estates (DESIGN TARGET): GPU nodes 1.8–2.4 Cr, object storage 0.8–1.2 Cr, bandwidth 0.6–1.0 Cr, twelve engineers 1.5 Cr; total about 5–6 Cr.

Hot / warm / cold recording of every frame is Model 4 and Phase-2, selected cameras only.

1.16 Test cases and experiments

The suite on 04 September 2026: 74 passed, 1 skipped (Tesseract binary). `scripts/audit_gate.py` prints PASS. `scripts/run_experiments.py -suite smoke` writes `05_Output/experiments/EXPERIMENT_LOG.md`.

What each family proves		
Family	Files	Claim
Integrator	<code>test_integrator_laws</code> , <code>test_scene_cut</code>	TCP, no FPS timing, scene-cut reset
Registry	<code>test_cameras</code> , <code>test_catalogue</code>	CSV, consent, catalogue 400 without host
ANPR	<code>test_plate_normaliser</code> , <code>negatives</code> , <code>honesty</code>	Confirm is not ANPR. Junk drops.
Objects	<code>test_objects</code>	Person blob inserts. YOLO without weights does not crash.
FRS	<code>test_faces</code> , <code>test_privacy_frs</code>	Same gallery matches. <code>cam04</code> emits zero faces.
Intrusion	<code>test_intrusion</code>	ROI overlap CRITICAL. Missing ROI is a no-op.
Matcher	<code>test_matcher</code>	Stolen, blacklist, <code>WL-004</code> without a plate, 120 s dedupe
Track	<code>test_track</code>	Six seed cameras. Append does not drop seed-1.
Security	<code>test_security</code> , <code>path jail</code> , no RTSP leak	Viewer 403 on enroll. Auditor 403 on confirm-face.

Experiment labels: E-A1 ANPR fixture (skipped without Tesseract), E-O1 person blob MEASURED, E-F1 gallery hit MEASURED, E-I1 intrusion MEASURED, E-W1 track count ≥ 6 MEASURED, E-S1 fifth session rejected MEASURED, E-S6 GPU 0 MEASURED. Government live grab (E-G*) SKIPPED when that process sees an empty `SENTINEL_HOST`.

1.17 How to run the PoC

Clone <https://github.com/amitduabits/PRAHARI>. From `02_Code/prahari` copy `.env.example` to `.env`. Set the judge password before any hosted demo. Set the Sentinel host, password, and RTSP IP after the



live portal login. Run `run.ps1` (Windows) or `run.sh` (Unix). Open `http://127.0.0.1:8080` and sign in as judge.

Three-minute walk: Operations map (seeded pins, Dahod offline), Cameras table Open tile on cam04, Vehicle Track reconstruct GJ01AB1234 and download CSV, Alerts CRITICAL and Ack, Onboard Analyse this still or confirm if OCR misses, Analytics & Gaps for short-retention Food and Civil Supplies cameras.

1.18 What is done, what is still human

Done in code and notes: hybrid plane, catalogue sync, RTSP-TCP frame, confirm CSV, objects, intrusion, lawful FRS with gov refuse, RBAC, experiment harness, this deck.

Still human before 12:00 IST on 07 September 2026: Tesseract on PATH if OCR should appear on camera; two consented adult photos for a live face demo; own-feed Unlisted YouTube ≤ 3 minutes; gov-feed Unlisted YouTube ≤ 3 minutes with no FRS; Drive Anyone+Viewer for the plate CSV; portal receipt.

